National University of Computer and Emerging Sciences, Lahore Campus

	Course:	**Data Warehousing**	**Course Code:**
	Program:	**BS**	**Semester:**
	Practice Problem2:	**Indexing Techniques - SOLUTION**	

Instruction/Notes:

Consider the following statistics:

Assumptions:

- Block Size = B = 50KB = 50 × 1024 = 51200 bytes
- Record Size = R = 256 bytes
- Total number of Records = r =1,280,000
- Available Memory= K = 100 blocks
- Index Record Size = *Ri* = 16 bytes

Query1:

*SELECT * FROM employee WHERE Department='Research' AND Gender='Female' AND Salary>300000;*

Query2:

*SELECT * FROM employee WHERE Department='Research' AND Gender='Female' AND Salary>50000;*

Now let's assume following selectivity for the query(s):

- Department="Research" has 55% selectivity
- Gender="Female" has 12% selectivity
- Salary >300000 has 4% selectivity
- Salary >50000 has 60% selectivity

Calculate the I/O cost for the above query(s) for all the indexes specified below.

1) FULL TABLE SCAN
2) SINGLE INDEXING
3) COMBINING MULTIPLE INDEXES
4) DYNAMIC BITMAP INDEX
5) STATIC BITMAP INDEX
6) COMPOSITE INDEX
7) CLUSTERED INDEX

Calculating measures necessary for calculations:
Blocking Factor = bfr = B/R = 51200 / 256 = 200
Blocking Factor for index= *bfri* = B/ *Ri* = 51200 / 16 = 3200
Blocks required for base table records = b = r / bfr = 1280000 / 200 = 6400
Blocks required for Index = *bi* = r/ *bfri* = 1280000 / 3200= 400

Answer for Query#1:
Combined Selectivity = Sc = 0.55 ×(0.12 ×(0.04 ×1280000)) = 3,380 rows
"Research" Selectivity = S1 = 0.55 × 1280000 = 704,000 rows
"Female" Selectivity = S2 = 0.12 × 1280000 = 153,600 rows
Salary greater than 300000 Selectivity = *S*3 = 0.04 × 1280000 = 51,200 rows

1. Full Table Scan:
Since we have to scan the Employee table for once, I/O cost is given as:
I/O cost = number of Base Table (Employee) Blocks = b = 6400 blocks

2. Single Indexing:
In this case we choose the highest selectivity index, which is salary> 300000. Its selectivity is given as:
Qualifying rows = S3 = 51200

Now since Qualifying rows > Base table blocks
i.e. 51200 > 6400
Hence, we have to read all the blocks of base table

Also, Index table access cost = *S*3 / bfri = 51200/3200 = 16 blocks
Total I/O cost = Base table access cost + Index table access cost
Total I/O cost = 6400 + 16 = 6416 blocks

3. Combining Multiple Indexes:
Since we are taking all indexes we will consider combined selectivity in this case:
Qualifying rows = Combined Selectivity = *Sc* = 3380 rows

Now since Qualifying rows < Base table blocks
i.e. 3380 < 6400
Hence, we have to read only 3380 blocks of base table

Also, Index access cost = Index1 access cost + Index2 access cost+ Index3 access cost
Total Index access cost = *S*1 / *bfri* + S2 / bfri + S3 / bfri
= (704000/ 3200) + (153600/ 3200) + (51200/ 3200)
= 220 + 48+16
= 284

Total I/O cost = Base table access cost +Total Index access cost

Total I/O cost = 3380 + 284 = 3664 blocks

Solution to the next questions: Do yourself – with appropriate assumptions/statistics, if required.

Answer for Query#2:

Combined Selectivity = Sc = 0.55 ×(0.12 ×(0.60 ×1280000)) = 50,688 rows
"Research" Selectivity = S1 = 0.55 × 1280000 = 704,000 rows
"Female" Selectivity = S2 = 0.12 × 1280000 = 153,600 rows
Salary greater than 50000 Selectivity = *S*3 = 0.60 × 1280000 = 768,000 rows

1. Full Table Scan:

Since we have to scan the Employee table for once, I/O cost is given as:
I/O cost = number of Base Table (Employee) Blocks = b = 6400 blocks

2. Single Indexing:

In this case we choose the highest selectivity index, which is gender=Female. Its selectivity is given as:
Qualifying rows = S2 = 153,600

Now since Qualifying rows > Base table blocks
i.e. 153,600 > 6400
Hence, we have to read all the blocks of base table

Also, Index table access cost = *S2* / bfri = 153600/3200 = 48 blocks
Total I/O cost = Base table access cost + Index table access cost
Total I/O cost = 6400 + 48 = 6448 blocks

3. Combining Multiple Indexes:

Since we are taking all indexes we will consider combined selectivity in this case:
Qualifying rows = Combined Selectivity = *Sc* = 50,688 rows

Now since Qualifying rows (i.e. combine selectivity) > Base table blocks
i.e. 50688 > 6400
Hence, we have to read all the blocks of base table

Also, Index access cost = Index1 access cost + Index2 access cost+ Index3 access cost
Total Index access cost = *S*1 / *bfri* + S2 / bfri + S3 / bfri
= (704000/ 3200) + (153600/ 3200) + (768000/ 3200)
= 220 + 48+240
= 508

Total I/O cost = Base table access cost +Total Index access cost

Total I/O cost = 6400 + 508 = 6908 blocks

Solution to the next questions: Do yourself – with appropriate assumptions/statistics, if required.
